SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY

(AUTONOMOUS)

Department Of Computer Science and Engineering

Insect Damage Detection in Household Surfaces Using AI-Based Image Processing

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Year & Sem : IV-II

ABSTRACT

The Insect Damage Detection System is an AI-based image processing platform that automatically detects and classifies insect damage on household surfaces such as wood, walls, and soil. Traditional manual inspection often fails to identify early-stage damage, leading to serious structural issues.

The system analyses images to detect patterns like holes, discoloration, cracks, and powder formation caused by insects such as termites and beetles. Using machine learning, it determines the severity of damage and provides treatment suggestions, preventive measures, and detailed reports, improving early detection and household safety.

EXISTING SYSTEM METHODS

- Manual inspection of household surfaces
- Visual identification of insect damage
- No automated monitoring system
- Dependence on human expertise

DISADVANTAGES OF EXISTING SYSTEM

- Time-consuming process
- High chances of human error
- Difficult to detect early-stage damage
- Lack of continuous monitoring
- Expensive maintenance after severe damage

PROPOSED SYSTEM

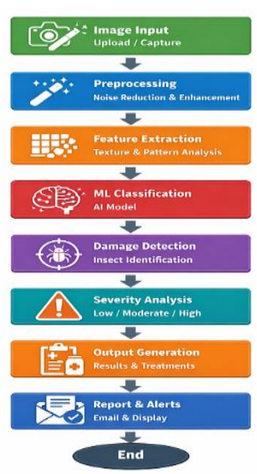
The proposed system introduces an AI-based image processing solution to automatically detect insect damage on household surfaces. It analyses captured images using machine learning techniques to identify damage patterns and classify severity levels.

The system provides accurate results along with treatment suggestions and preventive measures. It also generates reports and alerts for users, enabling early detection and reducing the risk of severe structural damage.

ADVANTAGES OF PROPOSED SYSTEM

1. Automated volunteer selection
2. Smart waitlist promotion
3. Attendance tracking
4. Absence approval workflow

FLOW CHART



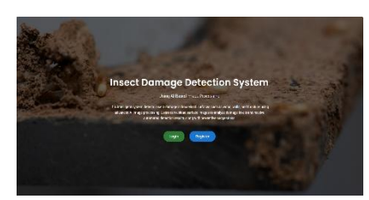
SDG Goals

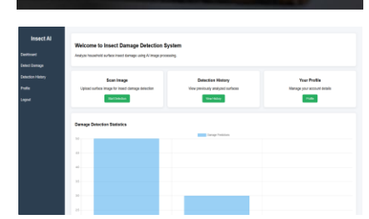
1. Sustainable Cities and Communities
2. Partnerships for the Goals

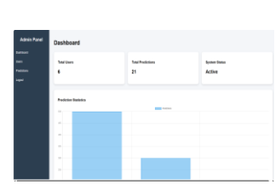
MODULES DESCRIPTION

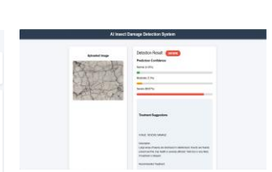
- **Image Processing Module:** Captures and preprocesses images for analysis
- **Detection Module:** Identifies insect damage patterns using AI algorithms
- **Classification Module:** Classifies damage type and severity
- **User Interface Module:** Displays results and suggestions to users
- **Notification Module:** Sends alerts and reports via email

OUTPUTS









CONCLUSION

The Insect Damage Detection System provides an efficient and intelligent solution for identifying insect-related damage in household environments. By utilizing AI-based image processing and machine learning techniques, the system enables early detection and accurate classification of damage patterns.

It reduces the need for manual inspection, minimizes human error, and helps users take timely preventive actions. The system also offers treatment suggestions and detailed reports, improving decision-making and maintenance planning. Overall, it enhances household safety, reduces repair costs, and serves as a reliable approach for smart and automated monitoring of insect damage.

BATCH NO: B – 10

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